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An Internet-based health gateway device for interactive communication and automatic data uploading: Clinical efficacy for type 2 diabetes in a multi-centre trial

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Abstract

Introduction: The aim of this study was to improve the quality of diabetes control and evaluate the efficacy of an Internet-based integrated healthcare system for diabetes management and safety.

Methods: We conducted a large-scale, multi-centre, randomized clinical trial involving 484 patients. Patients in the intervention group ($n = 244$) were treated with the Internet-based system for six months, while the control group ($n = 240$) received the usual outpatient management over the same period. HbA1c, blood chemistries, anthropometric parameters, and adverse events were assessed at the beginning of the study, after three months, and the end of the study.

Results: There were no initial significant differences between the groups with respect to demographics and clinical parameters. Upon six-month follow-up, HbA1c levels were significantly decreased from $7.86 \pm 0.69\%$ to $7.55 \pm 0.86\%$ within the intervention group ($p < 0.001$) compared to $7.81 \pm 0.66\%$ to $7.70 \pm 0.88\%$ within the control group. Postprandial glucose reduction was predominant. A subgroup with baseline HbA1c higher than 8% and good compliance achieved a reduction of HbA1c by $0.8 \pm 1.05\%$. Glucose control and waist circumference reduction were more effective in females and subjects older than 40 years of age. There were no adverse events associated with the intervention.

Discussion: This e-healthcare system was effective for glucose control and body composition improvement without associated adverse events in a multi-centre trial. This system may be effective in improving diabetes control in the general population.

Keywords

E-health, home telecare, online health

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Introduction

It is well established that regular blood glucose control is important for the prevention of diabetes and related complications.[1,2] Improving glucose control would reduce the enormous economic burden associated with the disease.[3,4] To this end, new information technology (IT)-based management systems for diabetes management have been developed and studied.[5–22] We established an interactive, internet-based communication system for patients and medical professionals and demonstrated its short-term efficacy for HbA1c reduction[5] and long-term efficacy for HbA1c reduction and glucose stability.[6]

Studies using IT are already increasing and their positive results are found all around the world. The blood glucose management system using IT is even focusing on commercialization which emphasizes a simple and convenient function for users. Convenient telecommunications

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systems for uploading glucose data were introduced and found to be effective for glucose monitoring and control.[23–25] Many studies are focusing on convenient automatic report systems that automatically input the blood glucose. However, these studies were performed with relatively small numbers of patients in only single-site trials, and were, therefore, not generalizable. Moreover, as most of the studies were conducted for research in hospitals, it is not easy for hospitals that have not experienced IT-based blood glucose management to apply and manage this system. Therefore, a standard guideline and process is needed to operate the IT-based blood glucose system, and there is still a lack of actual clinical examples.

In planning our study, we considered the difficulties encountered in previous studies, such as the fact that computer-based Internet use was found to be a barrier, especially for older patients, and that it was not convenient for patients to upload their glucose levels manually. Considering the importance of controlling blood pressure, lipids, and glucose levels to minimize the risk of diabetic complications, we planned to use a new Internet-integrated device which could upload patient data automatically. With three different hospitals, the study standardized the times of input of patients' blood glucose, medical staffs' medical feedback and the times of reporting, results of the blood test, etc. as one protocol.

Methods

Study population

We performed a six-month multi-centre, large-scale, randomized, prospective open trial to investigate of the clinical efficacy of an interactive, Internet-based communication and data management device. Men and women who had been diagnosed with type 2 diabetes for more than one year were recruited from the outpatient clinics of the Diabetes Center of Seoul St. Mary's Hospital, the Seoul Asan Hospital's Diabetes Center, and Kangbook Samsung Hospital's Diabetes Center from 1 July 2011 to 28 September 2012. All patients asked to participate had Internet access in their homes or offices and were trained to use the interactive communication system. Patients' baseline HbA1c levels ranged between 7.0% and 10.0%. Patients with any significant medical diseases were excluded. Those who had creatinine levels higher than 1.5 mg/dL for men (1.4 mg/dL for women) or aspartate aminotransferase (AST) or alanine aminotransferase (ALT) more than 2.5 times the upper normal limit were also excluded. We excluded patients who had not been taking stable diabetes medications for the three months before enrolment. Insulin pump users could not be included in the study. Patients who were participating in other clinical trials or plans were also excluded. The study protocol was reviewed and approved by the review board of each institution. Written informed consent was obtained from each participant. Of the 554 patients screened, 484 met all criteria and were enrolled for the trial.

Study design

Upon enrolment, participants were assigned to the intervention or control group using adaptive randomization[26] based on the Stratified Block Randomization Design: 244 to the intervention group and 240 to the control group. All participants were asked to visit each diabetes centre for baseline measurements of weight, height, waist circumference, and blood pressure. A fasting blood sample was obtained to measure the concentration of baseline HbA1c, blood sugar, blood urea nitrogen (BUN), creatinine, AST, ALT, total cholesterol, triglyceride, high-density lipoprotein (HDL)-cholesterol, and low-density lipoprotein (LDL)-cholesterol. Postprandial glucose level and urinalysis was also measured. Blood samples were taken to a central laboratory (Samkwang Medical Laboratories, Seoul, Korea) for HbA1c measurement, which was done by turbidimetric immunoassay. All participants were given a glucometer (OneTouch UltraEasy, LifeScan, CA, USA) with 150 strips and were asked to view a one-hour diabetes education program that included the method and frequency of self-monitoring blood glucose (SMBG), as well as nutrition and exercise information. All medications were recorded at the beginning of the study and at every medical visit. All participants were followed for six months and asked to visit the outpatient clinic every three months. At the three-month follow-up, they were asked to visit the diabetes centre for a medical interview with their physician. Participant blood samples were taken and blood pressure was measured. At the last follow-up at the end of six months, participants visited the diabetes centre again and the same laboratory tests performed at baseline were repeated. While the control group had received the conventional outpatient management during the study period, patients in the intervention group were taught how to use the Internet-based system and to use it during the study period.

Device and the intervention

The new device, a health gateway (HiCare, HX-461, Insung, South Korea) with Internet-based communication, to which a glucose meter and electronic manometer (UA-767PBT-C, AND) could be mechanically linked and data automatically transferred, was implemented within one week after randomization. Participants could upload the glucose and blood pressure data automatically to the online server through the device. Patients did not need to contact any specific website. Changes in weight and any questions or detailed information including diet, exercise, hypoglycaemic events, or other factors that could cause changes in the glucose level were also recorded by typing on the screen of the gateway device. Patients were able to see messages from the medical team with recommendations on the screen of the gateway device, as well as their laboratory data.

For our program, the staff consisted of two nurses and two diabetologists per institute. There was a separate

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centre for nutrition and exercise counselling, which included three dietitians and three exercise experts. The medical team provided consultation to patients who decided to receive additional individualized education for lifestyle management.

Nurses monitored the patient system, reviewed patient information and all uploaded glucose data, and sent optimized recommendations to patients based on the Korean Diabetes Management Guideline 2012 provided by the Korean Diabetes Association. The nurses mainly commented on lifestyle modifications. For requests that required physician approval, such as medication or dosage changes, nurses transferred patient requests to the physicians. Physicians reviewed patient information and sent their recommendations back to patients. The nurses transferred the patients to dieticians or exercise experts for more intense and individualized lifestyle modification.

We did not adopt an automated algorithm during this stage of the intervention. For the first three months, nurses sent a recommendation every week, and then, for the last three months, every other week. If patients in the intervention group did not upload their glucose data for more than one week, a warning message was sent. If patients continued to not record their glucose levels for more than three consecutive weeks despite such warning messages, they were withdrawn from the intervention group. Telephone-based communication was not used for follow-up and the participants were only contacted via the device's electronic chart system.

As conventional management in the outpatient clinic, patients in the control group visited the diabetes centre and were provided with recommendations about medication, medication dosage, and lifestyle modification from the diabetologist.

We defined compliance using a ratio of the number of uploaded SMBG data per recommended frequency of SMBG. We instructed patients to check their blood glucose levels at least twice per week when their HbA1c levels were within target range without any medication, at least once per day when their HbA1c levels were within target range with oral agents, more than twice per day when their HbA1c levels were within target range with insulin or when their HbA1c levels were outside of target range without any medicine, more than three times per day when their HbA1c were outside of target range with oral agents or insulin, or with multiple insulin injections, regardless of HbA1c levels. We defined good compliance as a ratio of more than 50% of their individually recommended SMBG frequency. In contrast, those uploading less than 50% of their recommended data were classified as in poor compliance.

Adverse events were coded using the Common Terminology Criteria for Adverse Events, Version 3.0, with relation to the intervention, and were assessed by the medical staff at follow-up visits or by patient self-reporting. We defined hypoglycaemia as less than 70 mg/dL of glucose concentration and events were assessed as adverse events.

The survey form (SF-12, a 12-item short-form health survey) was used to assess patient quality of life in five categories: activity, physical health, emotional problems, pain, and social activity, with four levels for each category. Using the survey, we obtained a physical component summary (PCS) and mental component summary (MCS). For evaluation of treatment satisfaction, we used the Diabetes Treatment Satisfaction Questionnaire Status (DTSQ), which comprised eight components.

Statistical analysis

All results are expressed as mean ± standard deviation. The data were analysed on an intent-to-treat basis, with the last observation carried forward used for the end point. To compare the two treatment groups, the independent *t*-test or Wilcoxon rank-sum test was used for continuous variables. For categorical variables, the Chi-square test or Fisher's exact test was used. Repeated measures ANCOVA was used to determine whether the secondary variables differed significantly over time or between the control and intervention groups. To compare within-group differences between baselines and follow-up, we used the paired *t*-test or the Wilcoxon signed-rank test. In the overall analysis, baseline measures with $P < 0.1$ between the two groups were used as covariant. For all tests, $P < 0.05$ was accepted as significant.

Results

Baseline characteristics

In this study, 484 local outpatients were randomized into an intervention ($n = 244$) or control group ($n = 240$), of whom 444 (91.7%) completed 24 weeks of treatment. The mean age of the control and intervention groups was 53.4 and 52.9 years, respectively ($p = 0.5519$). Participant clinical characteristics are shown in Table 1. Baseline demographic and clinical characteristics were generally the same in both groups. The proportion of male subjects was high in both groups, 152 (63.3%) and 155 (63.5%) in the control and intervention groups, respectively. Comparison of hematologic and blood chemistry results revealed no difference across groups. Both groups showed a similar pattern in medication for diabetes, hypertension, and dyslipidaemia.

Change of anthropometry and blood pressure

Anthropometric findings and blood pressure at baseline and follow-up are shown in Table 2. Body weight was reduced by 0.85 ± 2.47 kg ($p < 0.01$) in the control group and by 0.95 ± 1.96 kg ($p < 0.01$) in the intervention group, which was significant for intragroup change in both groups. Waist circumference was reduced significantly at three- and six-month follow-up by 0.92 ± 3.24 cm and 1.45 ± 2.93 cm, respectively, in the intervention group, which was a significant difference compared to the control group. For change

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Table 1. Clinical characteristics of subjects.

	Control	Intervention	*P*-value
Number of subjects	240	244	
Age, years	53.4 ± 8.7	52.9 ± 9.2	0.5519
<20, *n* (%)	1 (0.41%)	0 (0.00%)	
20 ⩽ age < 30, *n* (%)	1 (0.41%)	1 (0.42%)	
30 ⩽ age < 40, *n* (%)	18 (7.38%)	13 (5.42%)	
40 ⩽ age < 50, *n* (%)	63 (25.82%)	55 (22.92%)	
50 ⩽ age < 60, *n* (%)	110 (45.08%)	116 (48.33%)	
60 ⩽ age < 70, *n* (%)	44 (18.03%)	46 (19.17%)	
70 ⩽ age < 80, *n* (%)	7 (2.87%)	9 (3.75%)	
Sex (male), *n* (%)	152 (63.3%)	155 (63.5%)	0.9652
Weight, kg	69.7 ± 12.1	70.1 ± 12.5	0.6197
Height, cm	164.7 ± 9.1	165.1 ± 8.4	0.6160
Body mass index, kg/m^2	25.5 ± 3.2	25.6 ± 3.4	0.7980
Waist circumference, cm	88.4 ± 7.5	88.8 ± 8.7	0.6685
Systolic BP, mmHg	125.0 ± 12.6	125.0 ± 13.9	0.5527
Diastolic BP, mmHg	76.9 ± 9.4	77.2 ± 9.7	0.7970
Heart rate, bpm	79.0 ± 11.2	79.6 ± 11.0	0.5082
Diabetic complications, *n* (%)	121 (50.4%)	139 (57.0%)	0.1484
Retinopathy, *n* (%)	77 (43.0%)	105 (52.2%)	
Neuropathy, *n* (%)	52 (29.1%)	47 (23.4%)	
Nephropathy, *n* (%)	34 (19.0%)	30 (14.9%)	
Foot ulcers, *n* (%)	0 (0%)	3 (1.5%)	
Others, *n* (%)	16 (8.9%)	16 (8.0%)	
Anti-diabetes medication			0.6462
No medication, *n* (%)	4 (1.6%)	2 (0.8%)	
Single or dual combination, *n* (%)	78 (32.5%)	61 (25.0%)	
More than triple combination, *n* (%)	81 (33.8%)	96 (39.3%)	
Insulin ± oral agents, *n* (%)	77 (32.1%)	85 (34.8%)	
Anti-hyperlipide medication			0.9739
No medication	58 (24.2%)	53 (21.7%)	
Statin only	157 (65.4%)	161 (66.0%)	
Fenofibrate or Omege-3 only	6 (2.5%)	13 (5.3%)	
Statin and fenofibrate or omega-3 combination	19 (7.9%)	17 (7.0%)	
White blood cell, 10^9/L	7.17 ± 1.88	7.09 ± 1.86	0.5774
Haemoglobin, g/dL	14.2 ± 1.6	13.9 ± 1.7	0.2718
Platelet count, 10^9/L	252.7 ± 63.7	250.8 ± 68.1	0.5549
Fasting blood glucose, mg/dL	154.9 ± 33.4	157.7 ± 36.6	0.3992
HbA1c, %	7.81 ± 0.66	7.86 ± 0.69	0.4980
Total cholesterol, mg/dL	159.6 ± 35.1	157.6 ± 33.7	0.7291
Triglyceride, mg/dL	146.8 ± 78.2	151.6 ± 110.7	0.4149
HDL cholesterol, mg/dL	47.9 ± 9.9	47.5 ± 10.8	0.6873
LDL cholesterol, mg/dL	90.2 ± 30.1	88.2 ± 27.5	0.6988
AST, U/L	25.1 ± 9.7	26.4 ± 10.2	0.1712
ALT, U/L	29.3 ± 15.2	31.6 ± 17.7	0.3442
BUN, mg/dL	15.1 ± 4.1	14.7 ± 3.8	0.2840
Creatinine, mg/dL	0.84 ± 0.19	0.83 ± 0.21	0.3295
Calcium, mg/dL	9.10 ± 0.42	9.12 ± 0.42	0.5561
Phosphorus, mg/dL	3.60 ± 0.51	3.57 ± 0.52	0.4655
Sodium, mEa/L	140.5 ± 2.2	140.3 ± 2.21	0.2842
Potassium, mEq/L	4.36 ± 0.35	4.34 ± 0.39	0.6990
Total protein, g/dL	7.13 ± 0.38	7.10 ± 040	0.6497
Albumin, g/dL	4.37 ± 0.27	4.36 ± 0.29	0.9705
Total bilirubin, mg/dL	0.81 ± 0.30	0.82 ± 0.32	0.9922
Alkaline phosphatase, mg/dL	67.1 ± 20.6	65.5 ± 19.8	0.2186

Values are reported as mean ± standard deviation.
BP: blood pressure; AST: aspartate aminotransferase; ALT: alanine aminotransferase; BUN: blood urea nitrogen; HDL: high-density lipoprotein; LDL: low-density lipoprotein.

of systolic blood pressure, at six-month follow-up the intervention group showed a reduction of 2.90 ± 16.41 mmHg, which was significantly different, in contrast to a reduction of 0.76 ± 12.71 mmHg in the control group.

Effect on glycaemic control and lipid profile

For all subjects, change in fasting and postprandial glucose level, and lipid profile are shown in Table 2. HbA1c was significantly reduced by 0.31 ± 0.70% at six-month follow-up in the intervention group, while that of the control group was reduced by 0.11 ± 0.76%. There was significant difference between both groups at six-month follow-up ($p = 0.0102$). In ANOVA, there was a significant difference for interaction for time and group between both groups ($p = 0.0055$). In both groups, the fasting glucose level reduced significantly. However, the intervention group also showed a significant reduction in postprandial glucose levels at the six-month follow-up. We did not find any clinical difference in lipid profile in spite of a reduction in LDL-cholesterol level in both groups.

We analysed changes in HbA1c according to baseline HbA1c levels, compliance, sex, age, and smoking status (Figure 1) to investigate the effects of patient characteristics on glucose control. Patients with baseline HbA1c < 8.0% showed a significant decrease in HbA1c levels at six-month follow-up in the intervention group, while the control group showed an increased tendency. Interestingly, individuals below 40 years of age showed an increased tendency in HbA1c levels in both groups, while there was a significant difference between groups in those above 40 years of age and below 65 years of age. The intervention was more effective in women and non-smokers. Considering compliance and baseline HbA1c simultaneously, patients with good compliance and baseline HbA1 levels above 8% achieved a greater reduction of HbA1c by 0.8%, while patients with poor compliance showed similar results to the control group, despite the intervention. Patients with baseline HbA1c < 8% showed a decrease in both good and poor compliance. However, the control group showed an increased tendency at six-month follow-up. There was no significant change in weight and waist circumference in subjects younger than 40 years of age, while it was significant in subjects older than 40 in both groups. Waist circumference, especially, was more significant in the intervention group at the six-month follow-up.

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Table 2. Laboratory follow-up data of all subjects.

	Baseline		3 months		6 months	
	Control	Intervention	Control	Intervention	Control	Intervention
Weight	69.7 ± 12.1	70.1 ± 12.5	69.0 ± 12.0‡	69.5 ± 12.5‡	68.9 ± 11.9‡	69.1 ± 12.6‡
Δ *vs baseline*	–	–	−0.71 ± 2.29	−0.63 ± 1.73	−0.85 ± 2.47	−0.95 ± 1.96
Waist circumference	88.3 ± 7.9	89.0 ± 9.0	88.0 ± 8.0	88.1 ± 9.1‡	87.5 ± 8.1‡	87.5 ± 8.9‡
Δ *vs baseline*	–	–	−0.29 ± 2.55	−0.92 ± 3.24	−0.85 ± 2.86	−1.45 ± 2.93*
Body mass index	25.6 ± 3.3	25.6 ± 3.4	25.4 ± 3.2‡	25.4 ± 3.4‡	25.4 ± 3.4‡	25.2 ± 3.4‡
Δ *vs baseline*	–	–	−0.21 ± 1.09	−0.18 ± 0.73	−0.2 ± 1.28	−0.33 ± 0.77
Systolic BP	123.1 ± 12.6	125.0 ± 15.2	125.0 ± 12.8	123.1 ± 12.4	122.4 ± 12.8	122.1 ± 13.4†
Δ *vs baseline*	–	–	1.88 ± 13.6	−1.82 ± 14.2*	−0.76 ± 12.7	−2.90 ± 16.4
Diastolic BP	78.2 ± 9.2	79.0 ± 10.3	76.8 ± 9.7†	75.9 ± 9.1‡	75.5 ± 9.9‡	75.2 ± 9.2‡
Δ *vs baseline*	–	–	−1.43 ± 8.94	−3.12 ± 9.73	−2.74 ± 8.83	−3.83 ± 9.75
Fasting blood glucose, mg/dL	154.9 ± 33.4	157.7 ± 36.6	–	–	148. 1 ± 35.1‡	143.7 ± 34.2‡
Δ *vs baseline*	–	–	–	–	−6.86 ± 33.8	−14.0 ± 40.2
Postprandial glucose, mg/dL	224.8 ± 61.0	228.0 ± 64.2	–	–	223.1 ± 62.8	209.1 ± 60.0‡
Δ *vs baseline*	–	–	–	–	−1.65 ± 74.9	−18.6 ± 71.4*
HbA1c, %	7.81 ± 0.66	7.86 ± 0.69	7.70 ± 0.85‡	7.64 ± 0.83‡	7.70 ± 0.88‡	7.55 ± 0.86‡
Δ *vs baseline*	–	–	−0.11 ± 0.63	−0.22 ± 0.68	−0.11 ± 0.76	−0.31 ± 0.70*
Total cholesterol	159.6 ± 35.1	157.6 ± 33.7	–	–	155.7 ± 33.5	155.4 ± 32.9
Δ *vs baseline*	–	–	–	–	−3.84 ± 25.2†	−2.19 ± 25.9
Triglyceride	146.8 ± 78.2	151.6 ± 110.7	–	–	144.3 ± 82.4	145.4 ± 127.0
Δ *vs baseline*	–	–	–	–	−2.44 ± 68.3	−6.26 ± 128.5
HDL cholesterol	47.9 ± 9.9	47.5 ± 10.8	–	–	46.3 ± 11.0‡	46.0 ± 10.4†
Δ *vs baseline*	–	–	–	–	−1.61 ± 6.32	−1.48 ± 7.55
LDL cholesterol	90.2 ± 30.1	88.2 ± 27.5	–	–	84.9 ± 26.3‡	84.6 ± 26.0‡
Δ *vs baseline*	–	–	–	–	−5.32 ± 22.9	−3.59 ± 19.8

Values are reported as mean ± standard deviation.
BP: blood pressure; HDL: high-density lipoprotein; LDL: low-density lipoprotein.
*Comparison between control and intervention, Wilcoxon rank-sum test: * < 0.05
†, ‡Intragroup comparison, Wilcoxon signed-rank test: † < 0.05; ‡ < 0.01.

At six-month follow-up, women in the intervention group showed a significant reduction in waist circumference compared to baseline, while women in the control group did not. Moreover, waist circumference was reduced significantly in subjects with baseline HbA1c below 8% in both groups, while those with baseline HbA1c above 8% did not show any significant reduction in either group (Table 3).

Change of medication

We followed up on patient medications during the study period and there was no difference in medication between groups. While 10.2% of the intervention group and 10.8% of the control group with insulin ± oral agents continued to the end of follow-up, 15.7% and 15.6% of the patients stopped oral agents, respectively. In addition, 5.5% and 4.3% of the patients in the intervention and control groups, respectively, stopped insulin. Of the patients taking oral agents, 43.6% of the intervention group and 42.0% of the control group stopped, while 21.2% and 22.9%, respectively, of the patients maintained use of the oral agents during the study period. Only 2.5% and 1.7% of the intervention and control group taking oral agents started insulin during the study. As a result, 64.8% of the intervention group and 61.9% of the control group reduced their medication during the study period, which was not significant statistically.

Quality of life, treatment satisfaction, and adverse events

We evaluated changes in PCS and MCS to investigate patient quality of life. To measure treatment satisfaction, we used the DTSQ sum score. There was no significant difference in PCS and MCS between groups (Table 4), which suggests that the intervention did not decrease patient quality of life. DTSQ scores were significantly higher in the intervention group compared to that of the control group (27.9 ± 6.48 vs. 26.7 ± 5.8, $p < 0.05$), indicating that treatment satisfaction was higher with the intervention. Seventeen patients (6.9%) in the intervention

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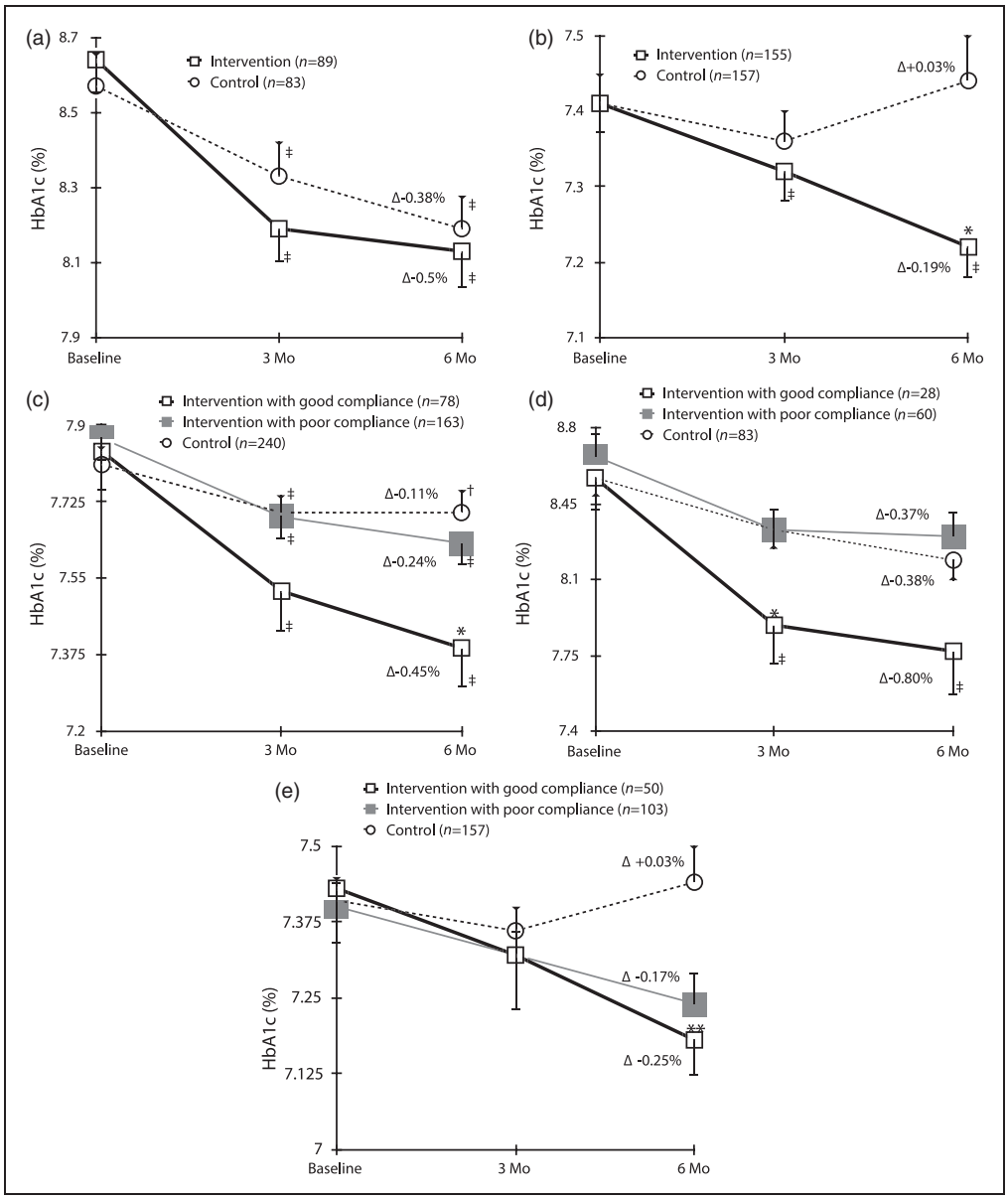

Figure 1. Subgroup analysis for HbA1c change during the study period.
Parts (a) and (b) show HbA1c change according to baseline HbA1c level; part (c), according to compliance; parts (d) and (e), according to compliance and baseline HbA1c; parts (f), (g), and (h), according to age; parts (i) and (j), according to sex; and part (k), according to smoking.
***Comparison between control and intervention: * < 0.05; ** < 0.01.
†‡Intragroup comparison: † < 0.05; ‡ < 0.01.

group and 15 (6.3%) in the control group reported all kinds of symptoms, which were classified into the adverse event. However, there was no side effect associated with the intervention system and no significance between groups ($p = 0.5127$).

Discussion

This study was a six-month multi-centre, large-scale, randomized, prospective open trial to investigate of the clinical efficacy of an interactive, Internet-based communication and data management device. In the study, we found that the e-healthcare system was effective for glucose control as well as for anthropometric parameter improvement compared to a conventional outpatient management system. We found significant HbA1c reduction with intervention of 0.31% compared to 0.11% in the control group of. Postprandial glucose reduction was especially predominant in the intervention group (−18.6 mg/dL vs. −1.65 mg/dL for the control group). The postprandial glucose reduction was also reported in a previous study using a mobile-phone-based communication system.[24] In addition, reduction of waist circumference was significant in the intervention group (−1.45 cm vs. −0.85 cm in the control group). There have been previous reports[5–25,27,28] showing the feasibility and efficacy of

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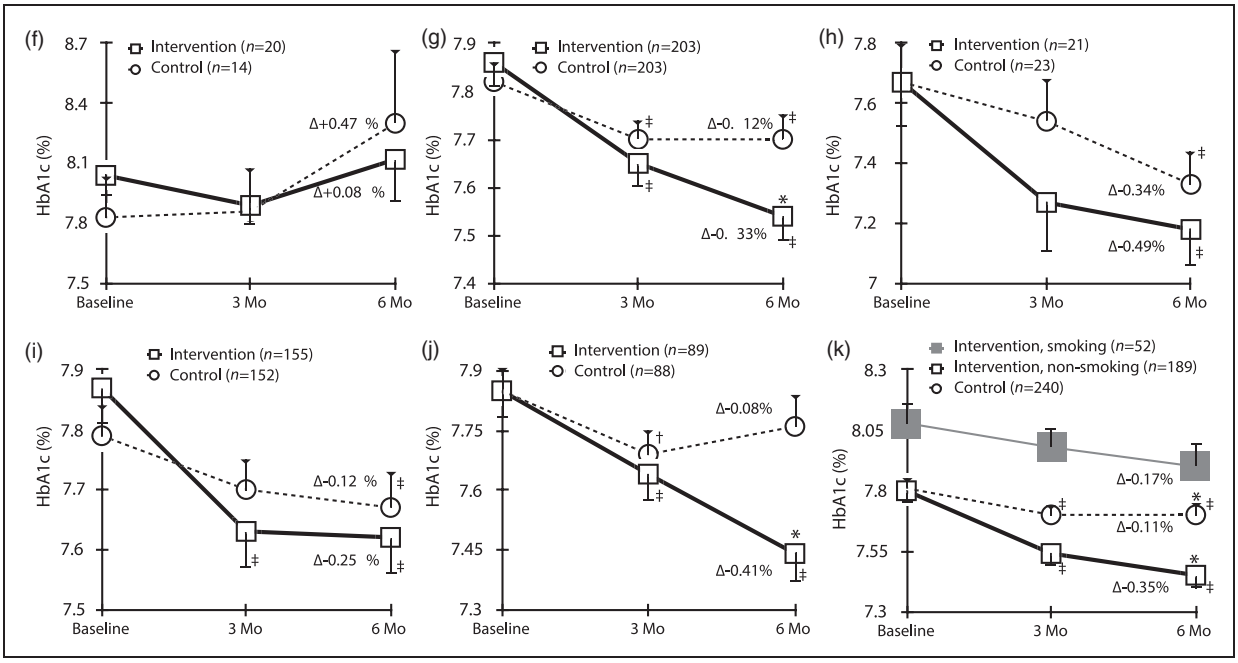

Figure 1. Continued.

IT-based communication systems for diabetes patients, but these studies were limited to small population groups in single centres. To be meaningful for general use, efficacy must be demonstrated in multiple centres and large populations. In this respect, the result of our study could be generally applied.

In subgroup analysis, patients with good compliance and baseline HbA1c above 8.0% showed about 0.8% HbA1c reduction with the intervention, while patients with poor compliance showed 0.38% reduction, the same as that of the control group (0.37%). This suggests that we could expect greater efficacy if we could improve compliance. Interestingly, subjects with baseline HbA1c below 8.0% showed a similar reduction of HbA1c in the intervention group regardless of compliance, while the control group showed an increase in HbA1c at six-month follow-up, which suggests that such a system could be effective for maintaining HbA1c stability for relatively well-controlled subjects. In addition, efficacy was more predominant in women than in men, and in patients older than 40. Waist circumference and weight reduction was also more effective in people older than 40 and women with the intervention compared to a conventional management system.

The most important thing in this system is the organic connection between device, platform (gateway) and medical feedback. In many previous studies, the auto-upload function of a glucometer was necessary for patient's attendance and convenience. Our study is mainly focusing on medical feedback as a large-scaled, multi-central, and randomized clinical trial. Unlike the previous studies, the strongest point of this study is that it used a standardized protocol applicable to three different large hospitals. Our study standardized the number of input of blood sugar, the medical staff's feedback, laboratory tests, etc. and applied the protocol to three hospitals at the same time. Therefore, any other hospitals that want to use this system but are not familiar with it can use the protocol easily. Therefore, the standardized protocol in device-platform (gateway)-medical feedback and the proven HbA1c lowering effect are the biggest strengths of this study.

We also followed up any adverse events to evaluate the system's safety. There was no difference in adverse events between groups and adverse events associated with the intervention were not observed. We did not observe a significant improvement in lipid profile and blood pressure with the intervention, but there was a slight reduction in systolic and diastolic blood pressure and LDL-cholesterol level. There was no difference between groups. This may indicate a limitation of the study. The study made use of three large diabetes treatment centres with diabetes experts and well-trained nurses on staff, so diabetes management, including blood pressure and lipid profile, was well monitored. For example, more than 50% of patients were taking anti-hypertensive medications and more than 65% of patients were taking anti-dyslipidaemia medications in both groups. Their baseline blood pressure and lipid profile was already within or very close to the target range. Average baseline HbA1c was also below 8.0%. We can assume that this system could show more efficacy in private clinics or small general hospitals where few diabetes experts are available. Another limitation of the study is that we could not differentiate compliance of the control group, because we could not monitor SMBGs, and, therefore, could not compare the intervention group to the control

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Table 3. Physical parameter follow-up data of all subjects.

		Weight		Waist circumference		Body mass index		Systolic blood pressure		Diastolic blood pressure	
		Control	Intervention	Control	Intervention	Control	Intervention	Control	Intervention	Control	Intervention
Age, years											
<40	Baseline	77.2 (16.1)	79.9 (17.7)	90.0 (9.2)	93.9 (13.6)	24.8 (3.6)	28.0 (5.4)	125.1 (13.4)	129.9 (13.1)	82.5 (10.9)	87.0 (14.0)
	3 months	76.8 (15.9)	79.6 (17.1)	89.3 (9.4)	93.5 (13.1)	24.7 (3.7)	27.9 (5.2)	122.2 (11.2)	125.9 (13.3)	79.1 (10.2)	81.5 (11.6)†
	6 months	76.0 (16.1)	79.5 (17.9)	88.6 (9.4)	93.5 (13.7)	24.6 (3.8)	27.9 (5.4)	121.1 (11.2)	126.4 (11.2)	78.8 (8.1)	81.5 (12.1)
40 and <65	Baseline	70.0 (11.7)	70.0 (11.8)	88.3 (8.0)	88.3 (8.0)	25.7 (3.3)	25.4 (3.2)	123.0 (12.8)	123.9 (15.6)	78.6 (8.9)	78.6 (9.6)
	3 months	69.3 (11.6)‡	69.1 (11.9)‡	88.0 (8.1)	87.4 (8.7)‡	25.5 (3.2)‡	25.2 (3.1)‡	125.2 (12.9)	122.9 (12.7)	77.3 (9.6)†	75.6 (8.5)‡
	6 months	69.2 (11.6)‡	68.6 (11.9)‡	87.6 (8.3)‡	87.2 (8.5)‡*	25.5 (3.4)‡	25.1 (3.2)‡	124.2 (13.1)	122.5 (13.5)	76.4 (9.7)‡	75.5 (8.6)‡
65≦	Baseline	63.0 (9.2)	64.6 (7.5)	87.5 (6.1)	90.5 (5.8)	24.9 (2.3)	25.0 (2.0)	122.4 (10.8)	131.0 (11.5)	72.8 (8.2)	75.3 (9.2)
	3 months	62.1 (8.7)‡	63.7 (7.6)†	87.5 (5.7)	89.7 (5.8)	24.8 (2.5)	25.0 (2.5)	125.1 (13.3)	123.2 (9.2)‡	70.6 (8.5)	73.2 (10.9)
	6 months	62.2 (8.6)‡	63.3 (7.7)†	87.6 (5.9)	89.1 (5.5)†*	24.8 (2.6)	24.9 (2.5)	125.2 (15.2)	125.0 (11.5)	70.4 (8.7)	75.0 (10.8)
Sex											
Male	Baseline	74.9 (10.4)	74.8 (9.9)	90.0 (7.2)	91.0 (7.1)	25.9 (3.0)	25.9 (2.8)	123.5 (12.5)	126.9 (15.1)	79.7 (8.7)	81.1 (10.2)
	3 months	74.2 (10.3)‡	74.0 (10.1)‡	89.8 (7.4)	90.2 (7.4)‡	25.7 (2.9)†	25.7 (2.8)‡	125.8 (12.4)	124.4 (12.9)†**	78.6 (9.0)	77.3 (9.7)‡**
	6 months	74.0 (10.4)‡	73.9 (10.2)‡	89.4 (7.5)‡	89.9 (7.2)‡	25.8 (3.1)†	25.6 (2.8)‡	125.0 (12.4)	124.2 (13.8)	78.2 (8.9)†	77.5 (9.7)‡
Female	Baseline	60.8 (9.4)	61.9 (12.3)	85.4 (8.3)	85.5 (10.7)	25.1 (3.7)	25.0 (4.2)	122.5*12.8)	121.5 (15.0)	75.6 (9.4)	75.4 (9.3)
	3 months	60.2 (9.1)‡	61.5 (12.4)†	85.0 (8.2)	84.5 (10.5)‡*	24.8 (3.5)‡	24.9 (4.2)	123.6 (13.4)	120.9 (11.4)	73.6 (10.1)	73.4 (7.6)
	6 months	60.2 (9.2)‡	61.1 (12.5)‡	84.8 (8.4)	84.3 (10.5)‡	24.8 (3.5)‡	24.7 (4.3)‡	122.6 (14.4)	121.0 (11.9)	72.0 (9.6)‡	73.1 (7.6)†
Baseline HbA1c, %											
>8.0%	Baseline	70.8 (11.6)	70.9 (14.5)	88.9 (7.8)	89.9 (10.4)	26.2 (3.4)	25.7 (4.2)	123.6 (13.8)	126.0 (16.5)	79.2 (8.9)	78.7 (10.8)
	3 months	70.2 (11.5)‡	70.4 (14.4)‡	88.8 (8.2)	89.2 (10.3)	26.0 (3.3)‡	25.5 (4.0)	126.1 (11.9)	125.9 (13.5)	77.8 (10.2)	76.9 (9.9)
	6 months	70.0 (11.5)‡	70.3 (14.6)‡	88.5 (8.3)	89.0 (10.1)	26.1 (3.6)†	25.5 (4.1)†	124.5 (12.1)	124.3 (13.8)	76.1 (9.7)‡	75.6 (9.9)‡
<8.0%	Baseline	69.2 (12.4)	69.6 (11.2)	88.0 (7.9)	88.5 (7.9)	25.2 (3.2)	25.5 (2.9)	122.8 (11.9)	124.3 (14.5)	77.7 (9.3)	79.2 (9.9)
	3 months	68.4 (12.2)‡	68.9 (11.3)‡	87.6 (7.9)	87.4 (8.3)‡	25.0 (3.1)†	25.3 (2.9)‡	124.4 (13.2)	121.6 (11.5)†*	76.3 (9.5)†	75.3 (8.7)‡
	6 months	68.4 (12.2)‡	68.6 (11.4)‡*	87.3 (8.0)‡	87.2 (8.1)‡	25.0 (3.1)†	25.2 (3.0)‡	123.9 (13.7)	122.3 (12.8)	75.9 (9.7)†	76.1 (8.9)‡
Compliance											
		Good	Poor	Good	Poor	Good	Poor	Good	Poor	Good	Poor
	Baseline	66.3 (12.3)	71.9 (12.3)*	87.8 (9.7)	89.4 (8.6)	24.9 (3.9)	25.9 (3.1)	125.8 (16.5)	124.7 (14.7)	76.8 (9.9)	80.2 (10.2)*
	3 months	65.5 (12.2)‡	71.3 (12.4)‡	86.4 (10.4)‡*	88.7 (8.3)‡	24.7 (3.8)‡	25.7 (3.1)‡	124.0 (11.9)	122.8 (12.8)*	76.4 (8.6)	75.7 (9.4)‡**
	6 months	65.4 (12.4)‡	70.9 (12.5)‡	86.9 (10.1)‡*	88.2 (8.3)‡	24.6 (3.9)†	25.6 (3.2)‡	121.6 (11.9)	123.8 (13.8)	75.1 (8.7)	76.4 (9.5)‡**

*, *Comparison between control and intervention, Wilcoxon rank-sum test: * < 0.05; ** < 0.01.
Values are reported as mean (standard deviation).
†, ‡Intragroup comparison, Wilcoxon signed-rank test: † < 0.05; ‡ < 0.01.

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Table 4. Quality of life and treatment satisfaction.

	Baseline		6 months	
	Control	Intervention	Control	Intervention
Quality of life				
Physical composite score	47.1 ± 8.0	47.0 ± 8.2	47.1 ± 8.0	47.9 ± 7.9
Mental composite score	51.0 ± 9.8	50.8 ± 9.8	50.5 ± 9.5	50.1 ± 10.2
Treatment satisfaction				
Satisfaction with current treatment	4.45 ± 1.29	4.32 ± 1.36	4.64 ± 1.22†	4.84 ± 1.23‡
Convenience of current treatment	4.23 ± 1.40	4.05 ± 1.35	4.45 ± 1.26†	4.67 ± 1.33‡
Flexibility of current treatment	4.25 ± 1.28*	4.00 ± 1.29	4.24 ± 1.29	4.48 ± 1.34‡
Satisfaction with own understanding of diabetes	3.89 ± 1.39	3.85 ± 1.27	4.11 ± 1.26	4.19 ± 1.37‡
Recommend present treatment	4.53 ± 1.32	4.41 ± 1.44	4.61 ± 1.26	4.86 ± 1.34‡
Satisfied with continued present treatment	4.51 ± 1.21	4.34 ± 1.37	4.70 ± 1.18†	4.91 ± 1.27‡
DTSQ sum score	25.9 ± 5.9	25.0 ± 6.3	26.7 ± 5.8	27.9 ± 6.48*

Values are reported as mean ± standard deviation.
DTSQ: Diabetes Treatment Satisfaction Questionnaire Status.
*Comparison between control and intervention, Wilcoxon rank-sum test: * < 0.05
†, ‡Intragroup comparison, Wilcoxon signed-rank test: † < 0.05; ‡ < 0.01.

group according to each compliance level. Our study was not enrolled by age as it was difficult to control the number of patients by age in a multi-centre. Additional studies by age groups are required.

Despite these limitations, the study suggests that this type of system is effective in glucose control, including HbA1c reduction and postprandial glucose level, and body composition, including waist circumference. Future studies should include more compatible patients, such as women, poorly controlled patients, and older people, in this type of treatment. We should develop a more intensive system for subjects with poor compliance and people younger than 40. In conclusion, we demonstrated the efficacy of the e-healthcare system for diabetes management in a larger population in multi-centres and there were no significant adverse events associated with the system. The next step will be to expand the use of the system to the general population.

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